

How to use this pack. If a new chat/developer has to take over, give them this file (or the PDF) plus access to the GitHub repo and Cloudflare account. Section A is the checklist Bernie uses every week. Section B is how to go back. Section C is the complete rule book. Sections D–F are the technical rebuild.

A — WEEKLY TRIPLE-CHECK (Bernie's checklist, every Wednesday before the auditor Excel)

Work **alphabetically, one worker at a time**, on /admin/wages. For each worker:

- Open reviews** — the orange **N open reviews** pill must be **0** before the Excel is pulled. Every red/orange review needs one click (see C7 for what each means).
- Duplicates** — same day, two rows? Scan the day columns. Same date + same or overlapping hours = one is a duplicate → **Edit shift** → amount **R0** with the reason "duplicate of #..." (or **Delete draft** if it was never final-submitted).
- Already paid last week?** — any row marked **MISSED SHIFT – actual date ...** (a date before last Saturday): was that date already in last week's payroll? The **ALREADY PAID** review will say so with the paid row as proof. Only the **difference** is paid (C5).
- Place = rate** — every "Warehouse Team" row: does the venue / area / description name a place that is *not* the warehouse (FNB, Riverside, Pretoria, a stadium, a house/garden)? Then it is a **venue** (R95/h) or **garden** (R62,50/h), not warehouse. The system flags it (**WAREHOUSE CLAIMED BUT THE ENTRY NAMES "..."**); click WAREHOUSE / VENUE.
- Meeting deducted** — Mon–Fri warehouse rows that start at or before 07:00 must show *"7,50 h paid (8,00 clocked)"* in the hours column. Applies on public holidays too. Petrus too.
- Sunday** — warehouse R97,50/h, venue R114/h (×1.2). Gardener House work R75/h (R62,50 × 1.2). Petrus Sunday is **held** until you approve.
- Public holiday** — every entry is a red review; approve at the holiday rule (C4). If you set everyone's hours for the day (e.g. "07–15 warehouse"), the next Monday's **follow-up panel** asks the real end time per person; **only underpayments** get a top-up.
- Green figure** — the green *Admin approved: N hours · R...* must equal what you intend to pay. Yellow = claimed, green = approved.
- Deductions & loans** — the **Fixed deductions** and **Additional loans** panels near the bottom: is what is due this week right? (Erick car R1 000 · Isaac eggs R260 · Joshua car R1 000 from 26 Sep · loan instalments per C9.)
- Pull the Excel** — **Download Payroll Excel (new)** (/wages-admin/payroll.xlsx). Check the **Auditor Summary** grand total against the dashboard total. Save a copy as docs/BW_Payroll_<from>_to_<to>_AUDITOR.xlsx. Send to the auditors.
- Stamp** — ask the assistant to write "WEEK APPROVED R..." on the worker rows (payroll note) and to make a restore point (git tag + JSON snapshot of wage_shifts, wage_payroll_reviews, loans/deductions).

Never: type a rate by hand · change a row in a closed (already-paid) payroll · pay anything twice for the same hours · pay Sunday ×1.5 · pay R375 half-days or R650/R750 fixed days to general staff.

B — GOING BACK TO A PREVIOUS PAYROLL / PREVIOUS VERSION

B1 — Just LOOK at a previous payroll (no change)

Dashboard: <https://bwprodssystem.co.za/admin/wages?from=2026-09-12> (any payroll Saturday). Excel: /wages-admin/payroll.xlsx?from=2026-09-12. Closed payrolls render read-only in effect — rule 7 says never change them.

Payroll Saturdays so far: 2026-08-01 ... 08-29 · **09-05** · **09-12** (paid 17 Sep, R71 096,90 approved) · **09-19** (auditors 23 Sep, R64 933,86 gross) · **09-26** (next).

B2 — Go back to a previous CODE version (look & feel / rules as they were)

```
git clone https://github.com/BWProductions/B-W-Costing.git webapp && cd webapp
npm install
git checkout <tag> # see table below
npm run build
npx wrangler pages deploy dist --project-name bw-productions --branch main --commit-dirty=true
curl https://bwprodssystem.co.za/wages-version # must print the version in the table
```

Deploying code **never changes data**. Rows, reviews, loans stay as they are.

Tag	Version	What you get
restore-2026-09-24-joshua-car	v2026-09-23-13	Current live. Everything in this pack.
restore-2026-09-23-payroll-final	v2026-09-23-13	Same code; data snapshot of the 19–25 Sep payroll as sent to auditors
restore-2026-09-23-difference	v2026-09-23-2	Difference-only WAREHOUSE/VENUE buttons (4-line sum, before the owner's single APPROVE wording)
restore-2026-09-23-reopen	v2026-09-22-16	Reopen button, area-aware crew check

Tag	Version	What you get
restore-2026-09-22-rules-v9	v2026-09-22-8	Hourly-by-place + Sunday ×1.2 + meeting; before crew check / holidays
restore-2026-09-22-rules-v8	v2026-09-22-7	Hourly-by-place (first version)
restore-2026-09-22-petrus-v6	v2026-09-22-6	Petrus R640 set day rule
restore-2026-09-21-already-paid	v2026-09-21-2	Already-paid rule; old Petrus rule; general staff still R750/R375 days
restore-2026-09-18-rolled-back	v2026-09-16-2	Before any already-paid / crew logic; Excel with editable HR cells
restore-2026-09-15-approved	v2026-09-15-x	Rate rules v4 + first Payroll Excel
weekly-wages-working-platform-baseline-2026-09-12	—	Baseline before this work started

B3 — Go back to previous DATA (undo a row/decision)

Every change session has a docs/restore-**<date>-<topic>**/ folder with ***_before.json** (the rows as they were). To undo, the assistant runs `UPDATE wage_shifts SET ... WHERE id = ...` from that JSON. Deleted drafts/rows are also copied into `wage_debug_capture / wage_deleted_shifts / wage_shifts_removed_audit`. Folders: `restore-2026-09-15-approved`, `-09-21-before-paid-before`, `-09-22-{petrus,rules-v8,stale-draft-720}`, `-09-23-{givemore,john,john2,joshua,holiday,final,loans}`, `-09-24-joshua-car`.

B4 — Undo the most common single things

Undo...	SQL the assistant runs
Petrus rent deduction removal	<code>UPDATE wage_recurring_deductions SET active=1, effective_to=NULL WHERE id=2</code>
Joshua car deduction	<code>UPDATE wage_recurring_deductions SET active=0, effective_to='<last day>' WHERE id=5</code>
A review decision	click  Reopen on the dashboard (no SQL)
A row amount	 Edit shift with a reason (writes <code>manager_update_reason</code> ; Excel shows "OWNER CORRECTION")
Planned end time for a day	<code>DELETE FROM wage_planned_hours WHERE work_date='2026-09-24'</code>

C — THE COMPLETE RULE BOOK (as live on 24 Sep 2026)

C1 — Payroll calendar

- Payroll week **Saturday** → **Friday**. Auditor Excel pulled **Wednesday**; Wed–Sun hours are best guesses, corrected by staff the following week (that is why the *Already paid* rule exists).
- Staff calendar allows **last Saturday** → **this Friday**. Older = "OLDER THAN ONE WEEK — cannot be final-submitted". Later = "NEXT PAYROLL — capture it from Saturday".
- A previous-week date captured now = **MISSED SHIFT**: real date kept, paid in the current payroll, pink pill, checked against last payroll.
- A draft is **R0** until Final Submission. After it: **Submitted — locked**.

C2 — General staff (hourly by place — rate rules v10)

Bhekizitha, Brian, Daniel, Erence, Erick, Isaac, John, Joshua, Patrick, Solomon, Thandanani, Thina (+ Givemore/Takavaudza when not doing House work; Lebo).

Place (from work type, cross-checked against venue/area/description)	Mon–Sat	Sunday ×1.2	Public holiday ×2
Warehouse	R81,25/h (R650 ÷ 8)	R97,50/h	R162,50/h
Venue / event (anything not warehouse)	R95/h	R114/h	R190/h

- Staff meeting 07:00–07:30 Mon–Fri unpaid** on warehouse entries (max 30 min = R40,63). Not on venue entries, not Saturday/Sunday. **Deducted on public holidays too** (deduct first, then ×2). Applies to Petrus too.
- Paid to the hour**, no rounding to days. Warehouse hours at R81,25 and venue hours at R95 are separate entries; travel gaps unpaid; overlapping entries counted once.
- Midnight split**: hours before 00:00 at that day's rate, after at the next day's (Sat 22:00–Sun 02:00 = 2×R95 + 2×R114 = R418).
- "Warehouse Team" only means warehouse when the PLACE says warehouse**. Venue/area/description naming FNB, Riverside, Pretoria, a stadium, botanical garden, a client's house → venue (or garden). Anything that is not Meyerton / Henley / Randvaal / the warehouse address is "a place". System flags; Bernie clicks.
- Examples: Wh Mon 06–16 = 9,5 h × 81,25 = **R771,88** · Wh Mon 07–16 = 8,5 h = **R690,63** · Wh Mon 08–16 = **R650** · Wh Sat 06–16 = **R812,50** · Wh Sun 06–14 = 8 × 97,50 = **R780** · Venue Mon 06–16 = **R950** · Venue Sun 08–16 = **R912** · Wh 06–12 + Venue 12–20 = R446,88 + R760 = **R1 206,88**.

Sunday ×1.2 and public holiday ×2 apply to EVERY place — confirmed by the owner 24 Sep 2026:

Place	Mon–Sat	Sunday (×1.2)	Public holiday (×2, held for approval)
Garden / House (gardeners)	R62,50/h	R75/h	R125/h

Place	Mon–Sat	Sunday (×1.2)	Public holiday (×2, held for approval)
Warehouse	R81,25/h (Mon–Fri meeting 07:00–07:30 unpaid)	R97,50/h	R162,50/h (Mon–Fri meeting deducted first)
Venue / event	R95/h	R114/h	R190/h
Petrus	R640 set day 06–16 (+R55/h held Mon–Fri, +R80/h Sat)	held: R640 + R80/h outside	R160/h every hour, meeting deducted
Lebo (student) — any place	R50/h every clocked hour — no meeting deduction	R60/h	R100/h
Music Bus	R750 fixed 07–16 + R120/h outside	R120/h	R750 fixed 07–16 + R180/h outside
A public holiday that falls on a Sunday is priced as a public holiday (×2), not ×2 × 1.2.			

C3 — Special people

Who	Rule
Music Bus (work type, usually John)	Mon–Sat R750 fixed for 07–16 + R120/h outside · Sunday R120/h · Holiday R750 + R180/h outside. Sat 08–22 = R750 + 6×120 = R1 470.
Petrus (Tsotlego Petrus Malakoane, staff 4)	R640 set day for 06:00–16:00 Mon–Sat, whatever time he arrives. Mon–Fri outside 06–16: R55/h held (red review: approve / other rate / R0). Saturday outside: R80/h automatic . Sunday: whole entry HELD (recommend R640 + R80/h outside; approve / other / decline). Holiday: R160/h every hour, meeting deducted, no fixed day (06:30–14:00 = 7 h = R1 120). No deductions, no loans (rent R450 ended 18 Sep).
Gardeners (Givemore staff 1, Takavaudza staff 2)	House / House-Garden work R62,50/h Mon–Sat; Sunday ×1.2 = R75/h (owner 24 Sep). Holiday R125/h (held). Their Warehouse Team / Team Assistance work = general staff by place. Takavaudza Sun 20 Sep #9821 "Installing lights and garden work" → VENUE R684 (6 h × R114; owner 24 Sep — was R375 garden flat, then R450 garden Sunday).
Sharleen Ndlovu (staff 17)	Fixed weekly R3 000 (payroll_rule = fixed_weekly).
John Wed 16 Sep	one-off owner correction R105/h (historic).

🏠 **Lebo — paid on his own schedule (owner 24 Sep 2026, v2026-09-24-4).** Owner: *"Lebo works differently as he's a student ... keep showing me his wages from Thursday the 24th going forward with a Paid button so I can push it when he's been paid and it disappears. Sometimes weekly, other times fortnightly or monthly."* Lebo (staff 16) is registered in wage_own_schedule_staff (from 2026-09-24). The dashboard shows a blue **"Paid on his own schedule"** panel listing every Lebo shift from that date not yet marked paid — **across payroll weeks** — date, times, hours, place, rand, a ✓ **Paid** button per line and ✓ **Paid — all N shifts R...** for the lot, and **TOTAL STILL TO PAY**. Pressing Paid stamps the row (own_schedule_paid_at/by, payroll note "PAID to worker (own schedule) by ... on ...") and the line leaves the list. R0 rows with an undecided review show "held — decide the review first" with no button. Amounts are never changed here (still 🗑 Edit shift / reviews). Route /wages-admin/own-schedule-paid (accepts one id or a comma list). Lebo still appears in the normal per-worker sections and the auditor Excel as before. **Lebo's rate (owner 24 Sep, v2026-09-24-7): R50/h at any place, EVERY clocked hour — NO 07:00–07:30 meeting deduction** (owner: *"because the wages are so little ... his rate is far less than anyone else's"*). Sunday ×1.2 = **R60/h**, public holiday ×2 = **R100/h** (held for approval). ownerPayKind → student / student_warehouse for staff 16; STUDENT_HOURLY = 50. State on 24 Sep: Thu 24 Sep Heritage 07:00–13:25 (owner: 13:25 is right, the 11:00 confirmation withdrawn and correction line #9864 removed) 6,42 h × R100 = **R641,67** (no meeting deduction) · Fri 25 Sep venue 9 h × R50 = **R450** · Sat 26 Sep venue 06:00–13:00 (owner: 13:00 per the Saturday note, was 12:40) 7 h × R50 = **R350 = R1 441,67 unpaid** (rows were R0 / R855 / R633,33 under the general-staff rule before the correction). Backups docs/restore-2026-09-24-lebo/. **Pay date (owner 24 Sep, v2026-09-24-6): ONE payment on Tue 6 Oct 2026** for everything from 24 Sep (wage_own_schedule_staff.next_pay_date); the panel shows a green "ONE PAYMENT DUE TUE 6 OCT 2026" banner; every clocked hour is paid. To add another own-schedule worker: insert into wage_own_schedule_staff.

C4 — Public holidays

Official gov.za list + Election Day **Wed 4 Nov 2026** (in SA_PUBLIC_HOLIDAYS in src/index.tsx and table wage_public_holidays). 2026 left: 4 Nov · 16 Dec · 25 Dec · 26 Dec. 2027 loaded. * **Never paid automatically** — every entry is a red review: **Approve public holiday — R...** / Other amount / Decline R0. * Rates: Warehouse R162,50/h (meeting off first) · Venue R190/h · Petrus R160/h (meeting off) · Music Bus R750 + R180/h · Gardener R125/h. * **Owner sets the day** (e.g. Heritage Day: "everyone warehouse 07:00–15:00") → planned end stored in wage_planned_hours (2026-09-24 → 15:00) → any entry past it gets red 🕒 **CLAIMED PAST 15:00** → the following week's dashboard shows the **Public-holiday follow-up panel**: per person, what was paid, dropdown of real end times 11:00–18:00 (half-hourly) each labelled as *paid / UNDERPAID, top-up +R... I worked less, nothing recovered, Confirm*. * **Owner rule (23 Sep, verbatim): "Even if they worked less, I don't want anything to be done. I only want something to be done if we underpaid them."** Top-up row only; never a recovery; answer written on the row (*HOLIDAY HOURS CONFIRMED*). * Heritage Day 24 Sep 2026 as approved: 9 men 07–15 wh **R1 218,75** · Patrick 06:30 **R1 300** · Petrus 06:30–14:00 **R1 120** · Givemore garden 08–15 **R875** · John wh 07–12 R731,25 + Music Bus 12–18 R1 110.

📌 **Owner's note panel (v2026-09-24-3, owner 24 Sep).** Any planned end time set for a day in the CURRENT payroll week (wage_planned_hours) shows at the top of the dashboard as a yellow note — *"Sat 26 Sep 2026 — work ended 13:00 (everyone at the warehouse ...)"* — with a live count of entries in so far and how many claim past the time. Entries past it carry the red 🕒 CLAIMED PAST pill. To add a note for a day, the assistant inserts a row into wage_planned_hours (work_date, planned_end, note, set_by). First use: Sat 26 Sep 2026 → 13:00. Sun 27 Sep 2026 → 11:40 (owner: everyone finished 11:40).

📌 **Owner DAY RULES (owner 27 Sep 2026, v2026-09-27-1).** Table wage_owner_day_rules(work_date PK, force_kind, staff_off_json, note, set_by). Two things the owner can set for a date: 1. **Forced place** — e.g. Sun 27 Sep: *"all the staff that select warehouse today will use the venue rate — they're building and fixing for a venue event."* force_kind = 'event' → every Warehouse Team / warehouse-wording entry that day is priced as **venue** (R95/h; Sunday R114/h) at Final Submission, on the dashboard green figure and in the Excel; the row's note says "OWNER DAY RULE ...". (force_kind = 'warehouse' does the reverse.) 2. **Staff OFF** — e.g. *"Daniel, Thina and Patrick are off — not allowed to claim Sunday."* staff_off_json = [10,3,13]. An entry from them on that date is **held at R0** at Final Submission with a red review 🕒 **You marked X OFF on this day ... If he did work: ... = R...** and buttons **He did work — approve R...** / Other amount / **He was OFF — decline, R0** (route /wages-admin/petrus-extra, key staff_off|shift:ID). 3. **Per-person START and END time** (v2026-09-27-2l-3; staff_start_json + staff_end_json, e.g. Takavaudza Sun 27 Sep 12:30–14:14 from the Life360 "left 52 The Avenue

HOUSE at 14:14" alert) — e.g. "Takavaudza's Sunday garden shift: he only arrived at 12:30 — can only claim from 12:30 onwards." staff_start_json = {"2": "12:30"}. If he enters an earlier start, the row is **priced from the owner's time** (his clocked times stay on the row; note "OWNER START ... paid from 12:30 only"; manager_update_reason set so the green figure and Excel keep it) and a red review "🕒 **You recorded that X only arrived at 12:30 ... paid from 12:30 (R...).** If he really was there from 10:00 the earlier hours would add R..." offers **OVERRIDE — he was there, add R...** / Other amount / **Keep 12:30 start — nothing more** (key owner_start|shift:ID). Live-tested on the preview (Taka 10:00–15:00 garden Sunday → 2,5 h × R75 = R187,50; override → R375), test rows removed. The 📌 **Owner's notes for this week** panel at the top of the dashboard lists each day rule (forced place, names off, "△ N of them entered a shift anyway") together with the planned-end notes. First use: **Sun 27 Sep 2026 — venue rate for all; Daniel, Thina, Patrick OFF**. Live-tested on the preview with temporary drafts (Daniel → held R0 + review; Bhekizitha warehouse → R684 venue Sunday), test rows removed.

C5 — Already paid → pay only the DIFFERENCE (rule of 21 Sep, wording of 23 Sep)

When a current-payroll entry overlaps a **paid** row from an earlier payroll, one red review with proof and one sum, in the owner's words:

Paid last week 08:00–16:00 · House/Garden · 8,00 h × R62,50 R500,00 (shift #9705, payroll 12–18 Sep) **06:00–18:00 at a venue 12,00 h × R95 R1 140,00 Less the Garden already paid – R500,00 WHAT YOU SHOULD PAY THIS WEEK R640,00 [✓ APPROVE R640,00 — VENUE R95/h]** (small: No — he was in the warehouse instead (R434,38))

Approve pays the difference **on this week's row** (last week's row never touched), writes "DIFFERENCE ONLY: ..." into the payroll note and Excel, voids the hour-based overlap reviews as superseded. Nothing due → R0. "Change the hours" fold-out re-prices live at the same rate; **Bernie never types a rate**. Guard rails: current unpaid payroll only, only against paid rows, dashboard/Excel only, never the staff app.

Garden entry whose wording sounds like EVENT work — the office is asked (owner 24 Sep 2026, v2026-09-24-2). Owner: "Installing lights in a garden — you should have asked me if that's a warehouse or a venue, because installing the lights could be a venue or garden work." Every gardener House / House-Garden entry (Givemore, Takavaudza) is now read for event words — lights/lighting, install, set-up, strike/breakdown, rig, stage, sound, décor, draping, marquee/tent, function, wedding, party, event, tables/chairs, dance floor, festoon/fairy — and flagged red **GARDEN CLAIMED BUT THE WORK SOUNDS LIKE A VENUE JOB** with two buttons: **GARDEN — R62,50/h (Sunday R75)** or **VENUE / FUNCTION — R95/h (Sunday R114)**, each showing the rand. Works on paid rows (re-prices now) and drafts (applied at Final Submission). First case: Takavaudza Sun 20 Sep #9821 "Installing lights and garden work" — **owner decided VENUE: R684,00** (6 h × R95 × 1.2). Paid R375 on 23 Sep → shortfall R309 corrected on the same row.

C6 — Duplicates / double-tap (v-11, 23 Sep)

- Phone lock:** first press greys all buttons → "Saving... please wait", full-screen overlay "Do not press again"; further presses swallowed; unlocks after 20 s.
- Save block:** new draft with same date + start + end as an existing draft or paid row is refused: "This shift (...) is ALREADY SAVED — it was not lost. Open it below and press Final Submission once." Editing an existing draft is never blocked.
- Final-submit block:** draft whose hours lie inside a paid row (this or an earlier payroll) is refused: "Not final-submitted: you have ALREADY been paid for ... Enter only the extra hours."
- Office catch:** Already-paid / overlap / crew reviews with the difference and one APPROVE; 🗑 **Delete draft**; 🕒 planned-end pill; stale drafts (before the window) in red with Delete.
- Excel = system:** R0 duplicates and approved differences appear identically in the Excel. Every refusal is logged in wage_debug_capture.

C7 — Reviews the office sees and what to click

Review	Colour	Click
ALREADY PAID — ... (paid_before)	red	APPROVE the difference / Nothing due R0 / change hours
WAREHOUSE CLAIMED BUT THE ENTRY NAMES "..." (crew_pattern, area-aware)	red	WAREHOUSE R81,25 / VENUE R95 (each shows the rand)
CREW PATTERN: 7 of 8 say Warehouse ... X says venue	orange	WAREHOUSE / VENUE
VENUE NOT NAMED	red	ask the worker; WAREHOUSE / VENUE
GARDEN CLAIMED BUT THE WORK SOUNDS LIKE A VENUE JOB (gardener entry mentions lights / install / set-up / strike / function ...) — owner 24 Sep	red	GARDEN R62,50 (Sun R75) / VENUE R95 (Sun R114)
Rate choice: Warehouse or Event/Venue ("warehouse"/"wearhouse" wording, other work type)	red	Warehouse / Event-Venue
Public holiday — ...	red	Approve R... / Other amount / Decline
Petrus: time outside 06–16 on a weekday / Petrus: Sunday entry	red	Approve R55/h · other rate · R0 / Approve Sunday · other · Decline
Manual overlap review (engine) — with "What overlaps what" proof box	orange	Approve only uncovered hours / Pay as claimed / No issue
Double claim — already paid (engine, catch-up > 1 week)	red	0 h
STALE DRAFT (before capture window)	red	🗑 Delete draft
🕒 CLAIMED PAST hh:mm	red pill	follow-up panel next week
Every decided review	green "Decided: ..."	🔄 Reopen if you need to change it

Self-compare reviews (draft vs its own paid row) are auto-voided. Decided flags never re-open by themselves; flags void themselves when the entry is deleted/corrected. **Pay-rule check panel retired 23 Sep** — do not rebuild it.

C8 — What the auditor sees (Excel, 6 tabs)

1 **Wage Detail Linked** (per shift: claimed/approved/effective hours & rand, rate, breakdown, review #, decision; yellow editable) · 2 **Auditor Trail Linked** (Sat...Fri HR/Amount, 6 missed shifts, hours, wages, fixed deductions due/deducted/outstanding, loans due/deducted/outstanding, total deducted, **NET WAGE**; formula-linked) · 3 **Auditor Summary** · 4 **Flagged – Reviewed** (every review, decision, who, when; per-worker pastel band, OPEN pink) · 5 **Loans & Deductions** (fixed deductions in force, every loan with balance and due-this-week, repayment history) · 6 **Missed Shifts**. Pricing precedence in the Excel: **owner correction on the row (manager_update_reason) is final** → RESOLVED rand decision (already-paid / difference) → RESOLVED approved hours priced by place → claimed. Fixed deduction counts in a week when effective_from <= weekEnd and (effective_to null or >= weekStart). Loan counts when active and deduction_start_date <= weekEnd, min(deduction_amount, outstanding).

C9 — Deductions & loans on the books (24 Sep 2026)

Fixed (weekly, ongoing): #3 Erick Car Payment R1 000 (from 1 Aug) · #1 Isaac Eggs R260 (from 1 Aug) · **#5 Joshua Car Payment R1 000 — from the 26 Sep payroll** (paid over to Joshua on the 25th monthly). Ended: #2 Petrus Rent R450 (18 Sep) · #4 Isaac Car R500 (23 Aug). **Additional loans:** #1 Erick R2 000 → R500 left, R500/wk · #2 Isaac R900 → R350 left, R250/wk · #3 Isaac R600 → R150 left · #6 Isaac R600 (kids' trip) → R600 due 19–25 Sep, left as scheduled · **#7 Erick R500** (22 Sep, asked R600) → R500 once, 26 Sep payroll · **#8 Isaac R500** fuel (22 Sep) → R250 on 26 Sep + R250 on 3 Oct · #5 Daniel R3 000 CT scan → **paid in full outside payroll**, no deductions. **Open question for Bernie:** Isaac's 26 Sep week would total R1 250 of loan deductions (+ R260 eggs) — cap or leave?

C10 — Standing rules (owner's, never to be broken)

1 Never break the staff flow — all checks on the office side. 2 Dashboard stays up while work is done. 3 Nothing is paid/unpaid/changed automatically — Bernie clicks. 4 A review must carry proof (the paid row: times, place, hours, rand, payroll). 5 Already paid → only the difference. 6 Rate comes from the place/work type; Bernie changes hours, never types a rate. 7 **Closed payrolls are closed** — corrections are a visible line in the current payroll. 8 Wednesday rule (estimates corrected next week). 9 Restore point + preview test with real cases + show Bernie before deploying; report old → new. 10 Keep the old CSV/Excel buttons until told. 11 Warehouse Team = warehouse only if the place says so. 12 Meeting deducted on holidays too. 13 Owner-set day → follow-up panel → only underpayments corrected. 14 One shift, one row. 15 An owner correction on a row is final.

D — TECHNICAL: WHAT THE SYSTEM IS

D1 — Architecture

- **Domain** <https://bwprodssystem.co.za> → Cloudflare Pages project **bw-productions** (branch main = production; branch realdate-test = preview).
- The project is a **Hono / TypeScript proxy** (`src/index.tsx`, ~5 800 lines). It sits in front of the original wages engine (UPSTREAM_ORIGIN = <https://3c3bcb89.bw-productions.pages.dev>, an older deployment of the same Pages project) and **rewrites its HTML with HTMLRewriter** to inject the new look, panels, buttons and rules, and adds its own routes. The engine still stores drafts/shifts and does the raw Final Submission; the proxy re-prices at submission (`applyOwnerRatesToFinalSubmission`) and writes the owner rules into the same D1 tables.
- **Database:** Cloudflare D1 **bw-productions-db** (id 4781960f-5bd6-4381-93fd-d8caf72f0c53), binding DB. R2 bucket `bw-productions-pdfs` (binding PDF_BUCKET, older PDF feature).
- **Stack:** Hono ^4, Vite 5 + `@hono/vite-cloudflare-pages`, Wrangler 3, TypeScript 5, `fflate` + `xlsx` (Excel written by `src/xlsx-lite.ts` / `payroll-excel.ts`). Frontend is server-rendered HTML + inline CSS/JS (no framework); gold `#c9a84c` accent, logo `public/static/bw-logo.png`.
- **Auth:** office cookie `bw_session = base64(json{id,name,email,role,exp}).hmacSHA256hex` with `ADMIN_SESSION_SECRET` in `index.tsx` (`adminUserFromCookie`). Worker session `bw_wage_session` (engine, name + PIN). Office can open a worker's draft via `/wages-admin/edit-draft/:id` (issues a 2 h worker session).

D2 — Source files

File	Purpose
<code>src/index.tsx</code>	proxy, all injectors, rate engine (ownerPayForShift, ownerPayKind, constants WAREHOUSE_HOURLY=81.25, VENUE_HOURLY=95, SUNDAY_FACTOR=1.2, MEETING_START/END=07:00/07:30, PETRUS_DAY=640, PETRUS_EXTRA_RATE=55, PETRUS_WEEKEND_EXTRA_RATE=80, HOLIDAY_FACTOR=2, PETRUS_HOLIDAY_HOURLY=160, MUSICBUS_HOLIDAY_EXTRA=180, SA_PUBLIC_HOLIDAYS), dashboard rendering (/admin/wages injector: reviews, pills, panels, green figure), all /wages-admin/* routes, double-tap protection, WAGES_UI_VERSION
<code>src/paid-before.ts</code>	Already-paid engine: rowToEntry, rowToPaid, priceAgainstPaid(deps, subject, paidRows, forceKind) → rate correction + extra hours + stillDue
<code>src/crew-pattern.ts</code>	Crew majority check, VENUE_WORDS, wordingNamesAVenue, venueEvidence, placeOf (→ 'ambiguous_area'), flag texts
<code>src/payroll-excel.ts</code>	6-tab auditor Excel, pricing precedence (owner correction final), fixed deductions & loans due
<code>src/xlsx-lite.ts</code>	minimal xlsx writer with styles/formulas
<code>docs/handover-2026-09-22-complete-rules.md/.pdf</code>	long-form handover (history 12–24 Sep, every decision)
<code>docs/REBUILD-PACK-2026-09-24.md/.pdf</code>	this file
<code>docs/restore-*/</code>	JSON before/after snapshots

File	Purpose
docs/BW_Payroll_2026-09-19_to_25_AUDITOR.xlsx	file sent to auditors

D3 — Routes added by the proxy

Route	Does
GET /wages-version · GET /wages-health	version string / health
GET /admin/wages?from=YYYY-MM-DD	office dashboard (proxied + injected)
GET /wages-admin/payroll.xlsx?from=	auditor Excel
POST /wages-admin/review-decision	approve hours / R0 / pay as claimed on a review
POST /wages-admin/rate-choice	WAREHOUSE / VENUE click (full day, or difference only when an earlier-payroll overlap exists)
POST /wages-admin/paid-before-decision	Already-paid: approve recommended / nothing due / changed hours
POST /wages-admin/petrus-extra · /wages-admin/holiday-decision	Petrus outside-hours / Sunday; public-holiday approve/other/decline
POST /wages-admin/reopen-review	↺ Reopen (keeps decisionHistory, reverses add-ons)
POST /wages-admin/delete-draft	🗑 Delete draft (copy to wage_debug_capture, voids reviews)
POST /wages-admin/confirm-holiday-hours	follow-up panel Confirm (top-up row only if diff > 0)
POST /wages-admin/manager-update (🔗 Edit shift)	owner correction on a paid row (manager_update_reason)
GET /wages-admin/edit-draft/:id	open a worker's draft as the office
Worker /wages/*	proxied; Save/Final Submission intercepted for the duplicate & already-paid blocks and re-pricing

D4 — Database tables (D1) that matter

wage_staff (id, display_name, payroll_rule hourly/fixed weekly, standard_weekly_amount) · wage_work_rates (staff/work type own rates e.g. House 62.5) · wage_shift_drafts · wage_shifts (paid rows: start/end, work_type, venue, area, description, amount, hourly_rate_snapshot, normal_hours, payroll_week_start, **manager_update_reason**, **payroll_note**, calculation_version, source_draft_id) · wage_payroll_reviews (issue_key unique e.g. paid_before|shift:ID, crew_pattern|draft:ID, holiday|shift:ID, petrus_extra|shift:ID; status OPEN/RESOLVED/VOID; approved_hours/amount; **system_snapshot_json** with placeDecision, differenceOnly, differenceText, approvedExtraAmount, decisionHistory, followUpNextWeek, ownerEndTime) · wage_planned_hours (work_date PK, planned_end, note, set_by) · wage_public_holidays · wage_recurring_deductions (fixed: type, amount, effective_from/to, active) · wage_weekly_deductions · wage_additional_loans (+ _repayments) · wage_debug_capture (every block/deletion logged) · wage_deleted_shifts, wage_shifts_removed_audit · wage_payroll_periods · wage_sessions, wage_login_attempts. Staff ids: Givemore 1 · Takavaudza 2 · Thina 3 · Petrus 4 · Bhekizitha 5 · Isaac 6 · John 7 · Erence 8 · Erick 9 · Daniel 10 · Solomon 11 · Brian 12 · Patrick 13 · Thandanani 14 · Joshua 15 · Lebo 16 · Sharleen 17.

D5 — Look & feel (so it can be recreated)

- Staff app: big white cards, gold accent #c9a84c, 19 px bold names, chevrons; one form per shift (date, start, end, work type, venue, area, description); **Final Shift Check** with real dates and tick boxes; *Submitted* — *locked* cards; *Finally submitted shifts* list; red banner messages for refusals.
- Office dashboard, top to bottom: header with **Download Payroll Excel (new)** + old export buttons · red **Public-holiday follow-up panel** (only the week after an owner-set holiday) · per-worker sections alphabetically: name, yellow claimed pill 59.00 hours · R5 190,00, green **Admin approved** pill (amber "N reviews still open"), orange **N open reviews** ▼, rows per day with pink **MISSED SHIFT**, red **REVIEW – ACTION NEEDED**, orange **REVIEW – open**, green **REVIEW – RESOLVED** + ↺ **Reopen**, red 🕒 **CLAIMED PAST**, grey hours column "7,50 h paid (8,00 clocked)", gold 🔗 **Edit shift / Open X's worker app** ↗, red **STALE DRAFT** + 🗑; review cells = white box with proof lines and big green APPROVE button · bottom: **Fixed deductions** and **Additional loan deductions to review** tables (read-only preview, grand total still to deduct).
- No Pay-rule check panel (retired).

E — REBUILD FROM SCRATCH (if the repo is lost too)

1. Create Cloudflare Pages project bw-productions bound to D1 bw-productions-db (existing data — never recreate the DB; if it is gone, the auditor Excel files in docs/ are the record of every paid payroll).
2. Hono + Vite Cloudflare Pages template; copy wrangler.jsonc from D1 above; npm i hono fflate exceljs.
3. Implement in this order, testing each on the preview branch with real rows: (a) proxy + HTMLRewriter shell and version endpoint → (b) rate engine C2–C4 with the constants in D2 and unit examples in C2/C4 → (c) Final Submission re-pricing + midnight split → (d) dashboard injector: pills, green figure, 🔗 Edit → (e) reviews framework (wage_payroll_reviews, issue_key, decision routes, Reopen) → (f) Already-paid engine (C5) → (g) crew/area check (C7) → (h) public holidays + planned hours + follow-up panel → (i) double-tap protection (C6) → (j) Excel (C8) → (k) deductions/loans panels.
4. Verify: npx tsc --noEmit clean of new errors; Excel vs DB 0 mismatches script; live test with temporary drafts, always deleted afterwards.
5. Tag, snapshot data JSON into docs/restore-..., update this pack.

F — WHERE THE COPIES ARE

- **GitHub:** <https://github.com/BWProductions/B-W-Costing> — main pushed 24 Sep 2026 with all tags (first push since 12 Sep; keep pushing after every session: `git push origin main --tags`).
- **Cloudflare:** production deployment history in the Pages project (each deploy is roll-back-able from the Cloudflare dashboard too).
- **Docs in the repo:** this pack, the long handover, auditor Excel files, restore JSONs.
- **Sent to Bernie:** PDF of this pack and of the handover (Genspark file links in chat).

Prepared for B&W Productions — Bernie Burness — 24 September 2026.

Owner notes ON EVERY ROW (v2026-09-27-6, 27 Sep 2026)

Bernie: "I asked you to show notes for me for Sundays... the warehouse doesn't show the message... it doesn't give me the times." Every owner note for a day is printed **on the row itself** (paid rows AND not-final-submitted drafts), not only in the panel at the top: -  forced place (e.g. "everyone at the VENUE rate today — this Warehouse entry is priced as venue, Sunday R114/h"); -  worker marked OFF that day (held at R0 with red review when final-submitted); -  per-person window (e.g. Taka 12:30–14:14): green "inside your window ✓", red "starts before / ends after" (trimmed to the window at final submission); -  entry ENTIRELY outside the per-person window (e.g. Taka draft 931 Warehouse 07:00–12:00 vs 12:30–14:14): **held at R0** at final submission with the staff-off style red review (approve if he did work / decline). Not merely trimmed. -  general finish time for the day (Sat 26 Sep 13:00, Sun 27 Sep 11:40): green within / red "x h past your time — ask what time he really finished". A per-person window overrides the general finish time for that person. Helper: `ownerNotesForRow()` in `src/index.tsx`; final-submission rule `outsideWindowNote` in `applyOwnerRatesToFinalSubmission`.

Several owner windows per person per day (v2026-09-27-7)

`wage_owner_day_rules.staff_windows_json = {{"staff_id": [{"start": "07:00", "end": "11:30", "place": "Warehouse"}, {"start": "12:30", "end": "14:14", "place": "House/Garden"}]}. At final submission an entry is paid for the piece inside a window only (red review to override); entirely outside all windows → held R0. Sun 27 Sep 2026: everyone left the warehouse at 11:30 (planned_end 11:30); Taka warehouse 07:00–11:30 + house 12:30–14:14. Lebo ledger: tick boxes + "Mark the TICKED shifts as PAID"; lines show NOT PAID until ticked. Lebo Heritage Day = 07:00–15:00 R800 (owner 27 Sep).`

Office edit form re-prices under the owner rules (v2026-09-28-1)

Bernie 28 Sep: "I'm trying to make changes. It's not allowing me." The engine's Edit Wage Shift form DID save the times, but re-priced the row with its old base formula ($R90/h \times \text{engine Saturday loading} = R840$ for Patrick #9869), so the owner rules were lost and it looked as if the change had not taken. Now after every accepted POST `/admin/wages/shifts/:id/edit` the proxy calls `repriceAfterManagerEdit()` (`src/index.tsx`): place from work type / wording / office decision / day rule, Sunday $\times 1.2$, holiday $\times 2$, warehouse meeting, student, gardener; stamps `calculation_version 10` and appends "Re-priced after office edit ..." to `payroll_note`. Rows carrying an owner decision (held at R0, owner-times cap, already-paid difference) are left as saved. The green message after Save now reads "... Priced under your rules: Venue/event: 7 h $\times$ R95 = R665,00 (the engine had put R840,00)."

Lebo Sunday $\times 1.5$ (v2026-09-28-2, owner 28 Sep 2026)

Lebo (student, staff 16) Sunday = $R50 \times 1.5 = \mathbf{R75/h}$ — Lebo only; the rest of the team stays on Sunday $\times 1.2$. `STUDENT_SUNDAY_FACTOR = 1.5` in `src/index.tsx`. Sun 27 Sep #9867 re-priced 4,5 h $\times$ R75 = R337,50 (was R270). Lebo ledger total due 6 Oct: R1 937,50. The day-rule "everyone at the venue rate" note on Lebo's rows says he stays on his student rate.

Owner amounts 28 Sep 2026 (v2026-09-28-4)

- Givemore #9794 Heritage Day Thu 24 Sep: owner set **R950,00** (formula gave $R875 = 7 \text{ h} \times R62,50 \times 2$). Owner amount — do not re-price.
- Thina (staff 3, fixed weekly) #9876: **R650,00** Heritage Day weekend allowance (owner 28 Sep). Real date 24 Sep, paid in payroll 26 Sep–2 Oct as a manager line with 0 hours, 00:00–00:00. Dashboard shows an "OWNER AMOUNT – for Thu 24 Sep" pill instead of the missed-shift / time checks (`isOwnerAmountLine: hours 0 and start == end` → no planned-end / owner-note checks).

"CATCH-UP" replaces "MISSED SHIFT" everywhere the owner sees it (v2026-09-28-5, owner 28 Sep 2026)

Bernie: "CATCH-UP — use this term, but you always need to explain it, because I can't read your mind." Definition shown with the label every time: a shift the worker did in a PREVIOUS payroll week (already paid out) but only entered afterwards; paid now, in the current payroll, showing the real date worked; checked against what was already paid that day (OVERLAP / CLEAR) so no hour is paid twice. "The day was not missed — the entry was late." Where: admin dashboard legend #bw-catchup-legend, CATCH-UP pill (hover text), per-row "What CATCH-UP means" line on paid rows and drafts, totals ("of which CATCH-UPS"), Excel (column header, sheet 6 title/headers), worker app locked-section heading and card text. Internal DB marker `missed_previous_week / description prefix "MISSED SHIFT - actual date ..."` is unchanged (the engine relies on it).

Petrus fixed weekly wage — R3 840 guarantee (v2026-09-28-7, owner 28 Sep 2026)

Owner: "It's a fixed day rate, Monday to Saturday. No changes unless I tell you. Public holidays are different, but it doesn't apply to Saturday and Sunday. If there's a public holiday on Saturday/Sunday we take 640×6 and put it in 5 days. You must always clear 3 840. We put it over 6 days for UIF and VAT purposes." - Constants `PETRUS_STAFF_ID = 4`, `PETRUS_DAY = 640`, `PETRUS_WEEK_DAYS = 6`, `PETRUS_WEEK_GUARANTEE = 3840`. - `ownerPayForShift`: for kind 'petrus' a Sat/Sun date is never treated as a public holiday (ordinary R640 Saturday; Sunday held as before). Mon–Fri public holiday keeps the holiday rule (hours – meeting $\times$ R160, held for approval) — that week clears MORE than R3 840. - Dashboard panel #bw-petrus-week: the 6 days Mon–Sat of the payroll week, each paid / draft / today / still to come / nothing in; "In this payroll so far" vs the guarantee (R3 840, or $R3\,840 - R640 + \text{holiday amount when a weekday holiday is priced}$). Button "Top up the missing day(s)" → POST `/wages-admin/petrus-guarantee` adds an owner line (event_name 'Petrus weekly guarantee', 0 h, R640) for each fully-past Mon–Sat day with no paid row and no draft. Only days before today count as missing. - Week 19–25 Sep check: $5 \times R640 + \text{Heritage Day } R1\,120 = R4\,320$ ✓.

No "Final Shift Check" page — FINAL SUBMISSION submits immediately (v2026-09-28-8, owner 28 Sep 2026)

Owner: "I've told you multiple times we don't want this anymore. When you push final submit, it just finally submits it." The worker's FINAL SUBMISSION button is a GET link to `/wages/drafts/:id/final-check` (engine's confirmation page with tick boxes). The proxy now treats that GET like the confirmation POST: rewrites it to POST `/final-submit` with the engine's three yes-answers (`time_correct / end_time_correct / information_complete`), runs the normal final-submit handling (Saturday parking, real date, owner-rule re-pricing) and redirects to `/wages/me?`

submitted=1. The check page is never shown. Errors go to /wages/me?error=.... Escape hatch for debugging only: ?bw_show_check=1 still renders the engine page. Tested 29 Sep with a throwaway draft (removed).

Owner corrections 29 Sep 2026 (v2026-09-29-1)

1. Holiday follow-up panel ("REALLY WORKED UNTIL") excludes owner-amount lines (0 h, start = end, e.g. Thina R650) and fixed-weekly staff — those were paid properly and need no re-check.
2. Petrus on a Mon–Fri public holiday: the owner MUST approve (was he actually there?). Review offers two buttons: "Approve public holiday rates — R..." (hours – meeting × R160) or "Approve fixed rate — R640,00" (choice fixed_day), plus other amount / R0.
3. Planned finish time (Sat 26 13:00, Sun 27 11:30 etc.) applies to WAREHOUSE and VENUE staff only — never to Music Bus entries (work type / venue contains "music bus") and never to Petrus. `planEndApplies()` gates both the CLAIMED PAST pill and the on-row note.

Lebo venue rate + Sonop 14:00 (v2026-09-29-3, owner 29 Sep 2026)

Lebo: R50/h is WAREHOUSE ONLY. Venue / collection / setup / event = R75/h (STUDENT_EVENT_HOURLY). Sunday R75/h anywhere. Holiday R100/h. Sat 26 Sep Sonop team (Solomon, Bhekizitha, Daniel, Lebo, Thandanani, Erence) worked to 14:00 — rows re-timed and re-priced; Patrick/Joshua were NOT at Sonop and stay 13:00. A per-person owner window overrides the general planned-end pill. Full detail: docs/restore-2026-09-29-sonop-lebo/.

Update 29 Sep (v2026-09-29-4)

- Daniel #9897 Sat 26: NOT on the Sonop team (only 07:00 starter; enters 07:00 on every shift) — reverted to 07:00–13:00 = R570 (owner option B). Sonop 14:00 team is: Solomon, Bhekizitha, Lebo, Thandanani, Erence.
- Lebo rate is read from WHAT HE WRITES (owner: "just read what he writes"): `studentKindFromWording()` — event words (collect / setup / deliver / drop-off / breakdown / strike / standby / activation / event / venue / golf / stadium / hotel / arena / festival / wedding / function / install / rig) → R75, and they WIN over the word "warehouse"; otherwise warehouse words (load / offload / prepar / pack / clean / wash / sort / stock / receiv / warehouse) or Warehouse work type → R50; anything else → R75. No owner question needed.